



DIGITAL VLSI DESIGN

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DIGITAL VLSI DESIGN

PART 2

Fabrication of MOSFETs with Layout Design

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Fabrication of MOSFETS with Layout Design

UNIT 1 (Part 2): Fabrication of MOSFETS with Layout Design

Introduction, Fabrication Process Flow: Basic Steps (2.1 & 2.2)

The CMOS n-Well Process (2.3)

Layout Design Rules (2.4)

Full-Custom Mask Layout Design (2.5)

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Fabrication of MOSFETs

Part a – NMOS Fabrication Process

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Fabrication of MOSFETs & Circuit Design Process

E-nMOS Fabrication Process

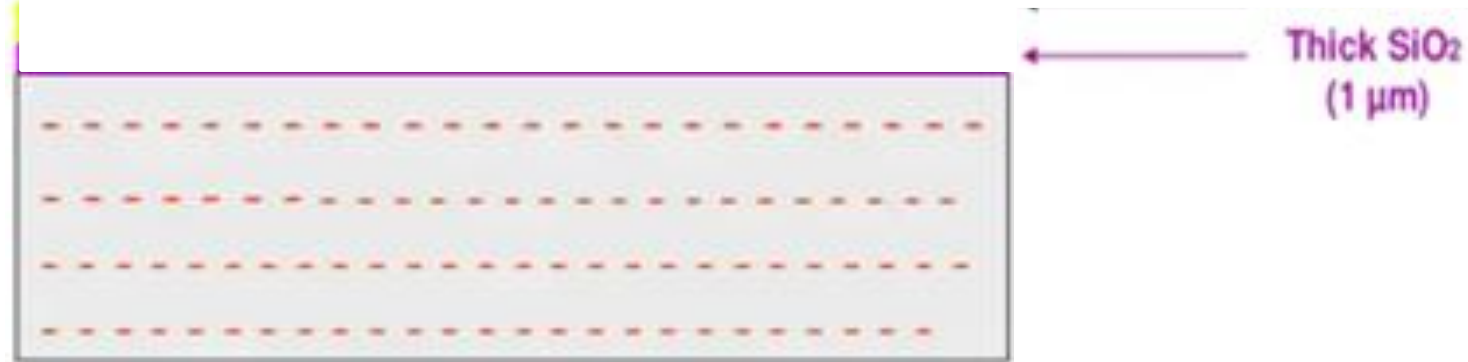
Pure Si

Si-substrate

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Doping of P-type Impurity to Pure Si Single Crystal



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nMOS Fabrication Process

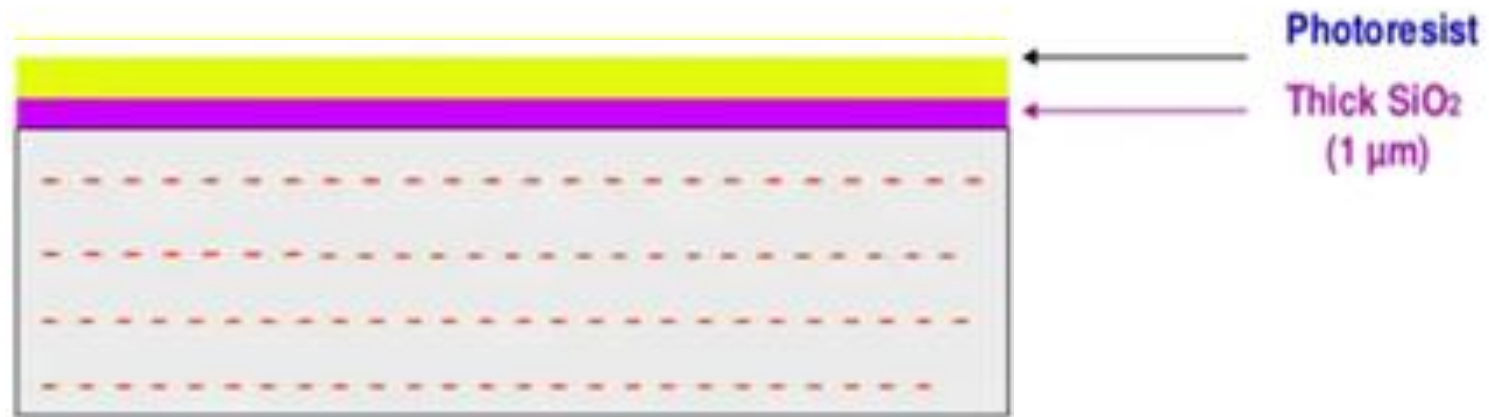
Deposition of SiO₂ layer over Si surface



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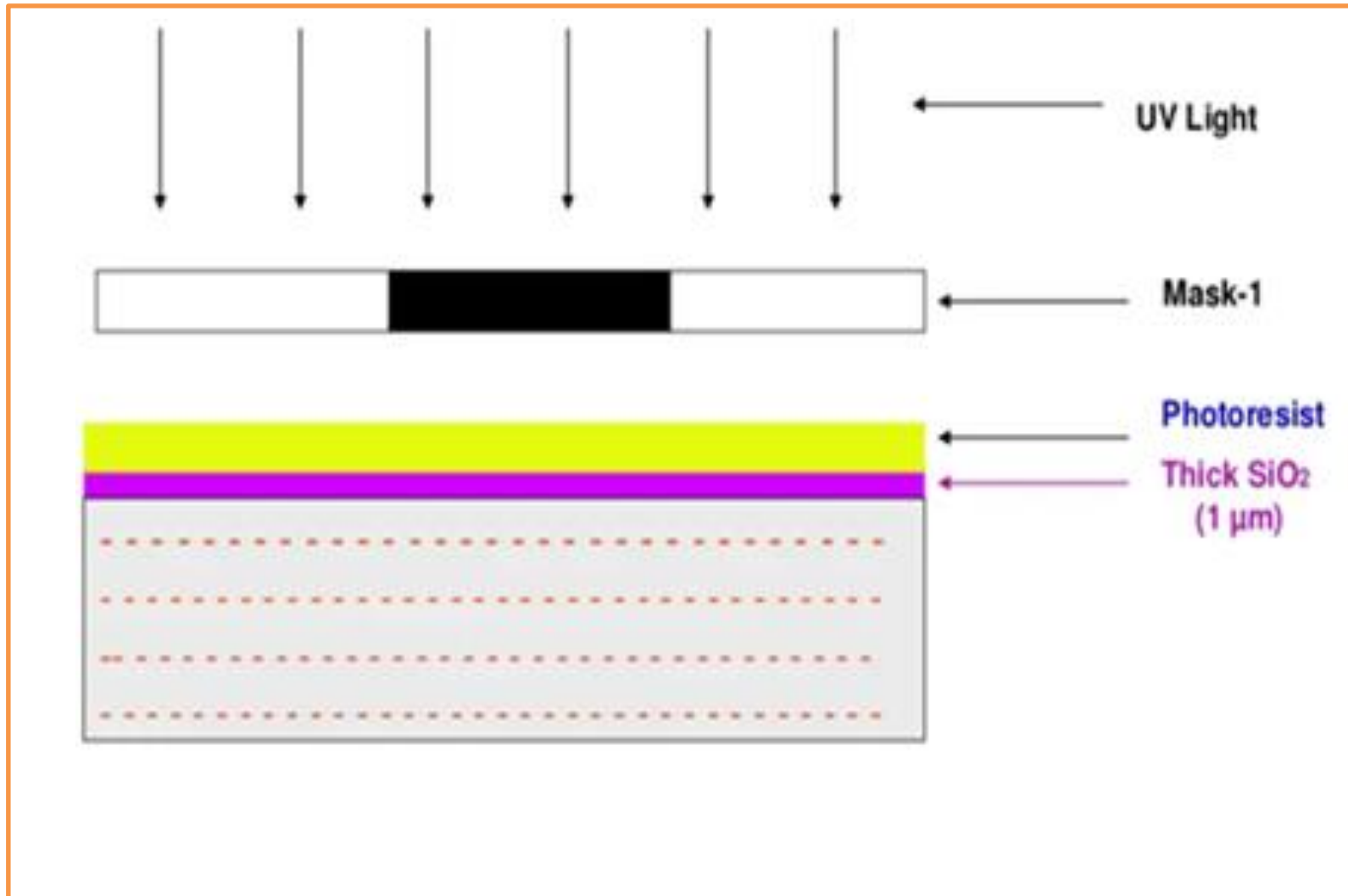
nMOS Fabrication Process

Photoresist Deposition over of SiO₂



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Photoresist Layer Exposed to UV Rays through MASK1

The entire SiO₂ is covered with a layer of photoresist, which is essentially a light-sensitive, acid-resistant organic polymer, initially insoluble in the developing solution

Mask1 is used to expose SiO₂ where S,D and G terminal to be formed. It is basically ACTIVE REGION where MOS device is fabricated

The photoresist exposed to UV light, makes it soluble & Not resistant to etching solvents.

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Etching soluble Photoresist (Positive Photoresist exposed to UV becomes soluble)

The MASK defines the area to be exposed to UV rays. Thus, when the structure with the mask on top is exposed to UV light, areas which are covered by the opaque features on the mask are shielded.

In the areas where the UV light can pass through, on the other hand, the photoresist is exposed and becomes soluble.



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Etching of SiO₂ which is exposed Etching agent for SiO₂



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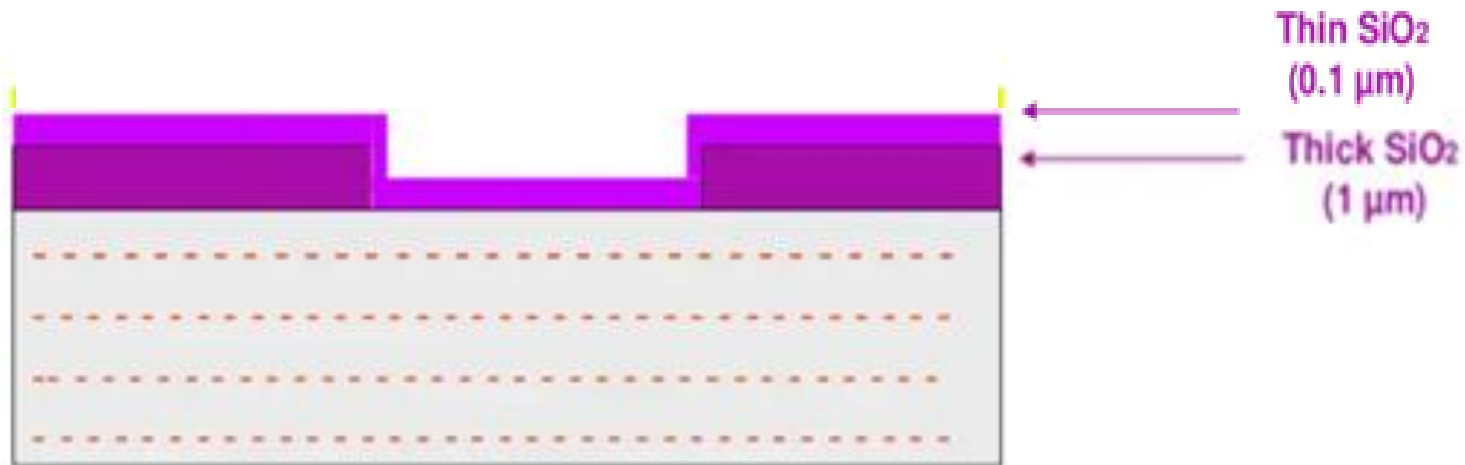
**Active AREA: Etching of
Hardened Photoresist Layer
using appropriate etching
agent (H_2SO_4)**



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**Formation of POLY gate:
Step1: Depositing Thin SiO₂
layer**



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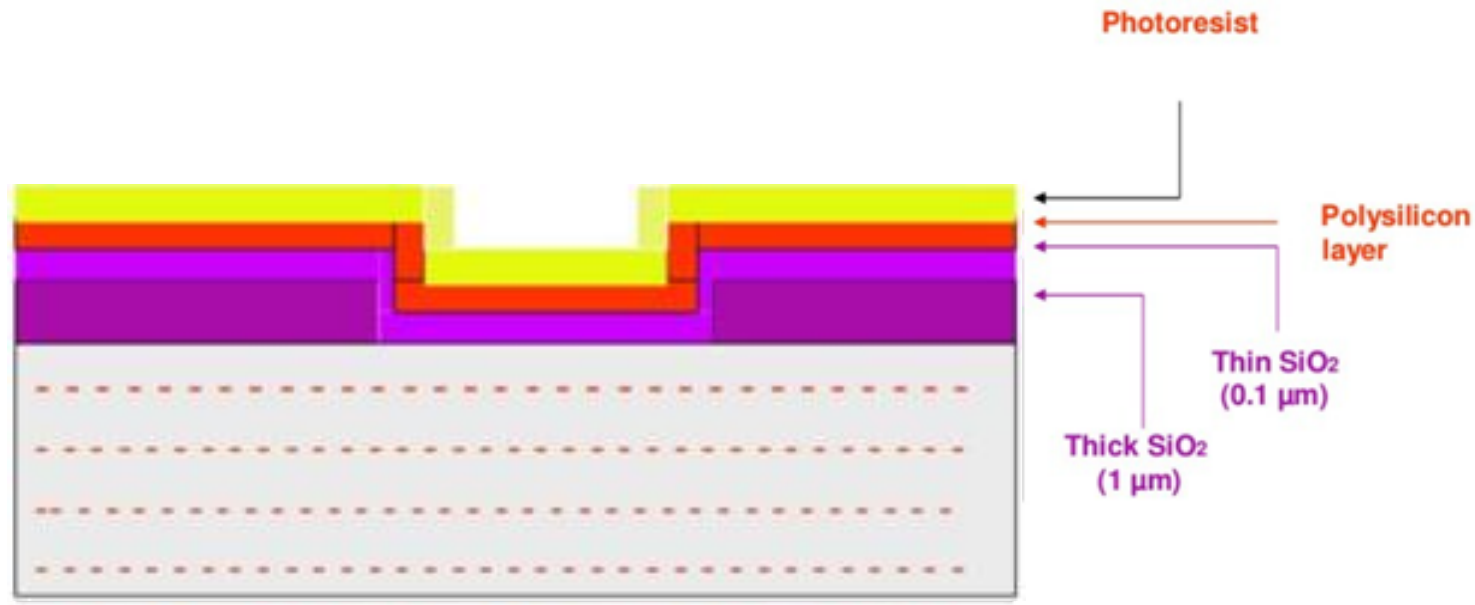
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Formation of POLY gate:
Step1: Depositing Thin SiO₂ layer
Step2: Depositing Polysilicon Layer



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Formation of POLY gate:

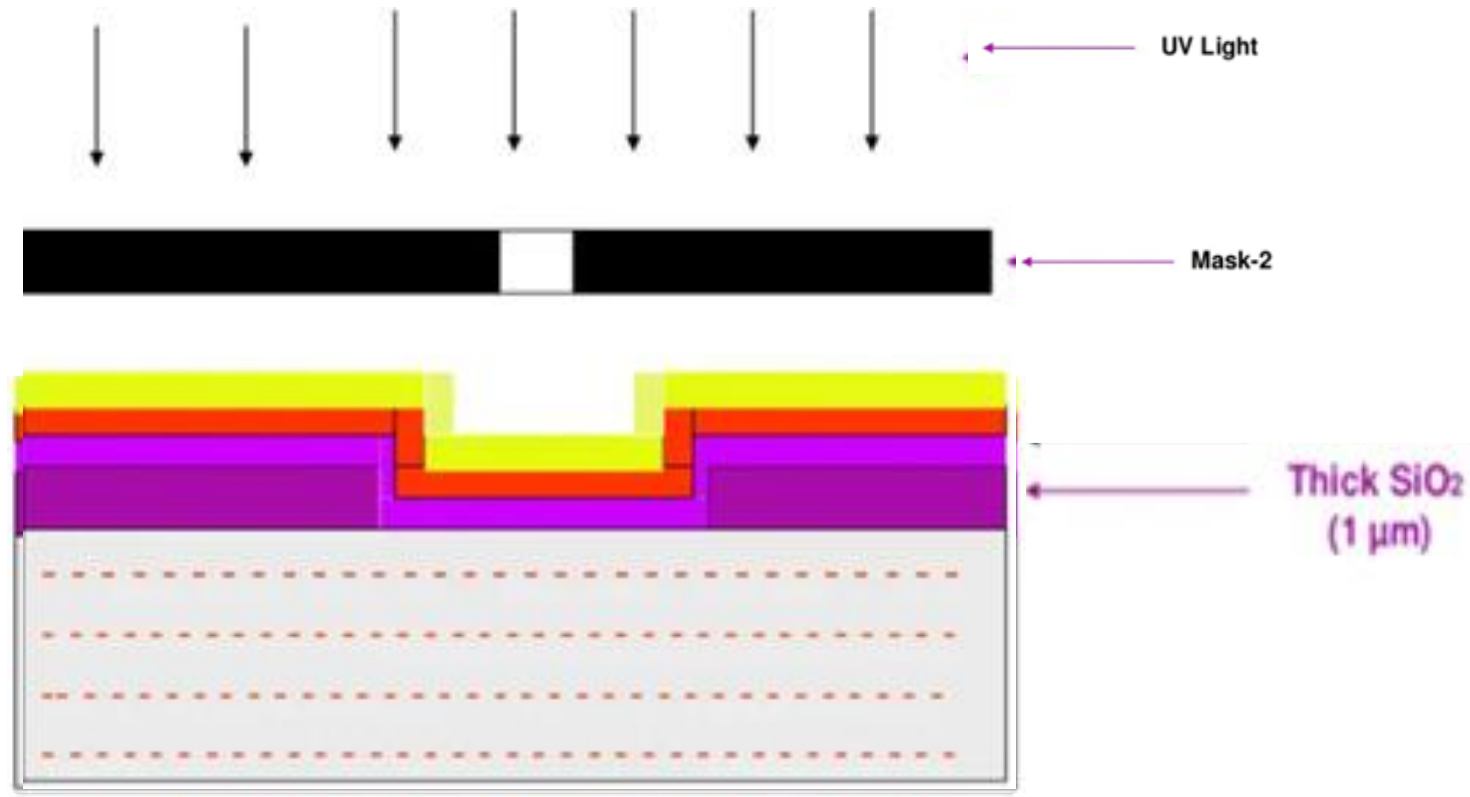
Step1: Depositing Thin SiO₂ layer

Step2: Depositing Polysilicon Layer

Step3: Deposition of Photoresist layer

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Formation of POLY gate:

Step1: Depositing Thin SiO_2 layer

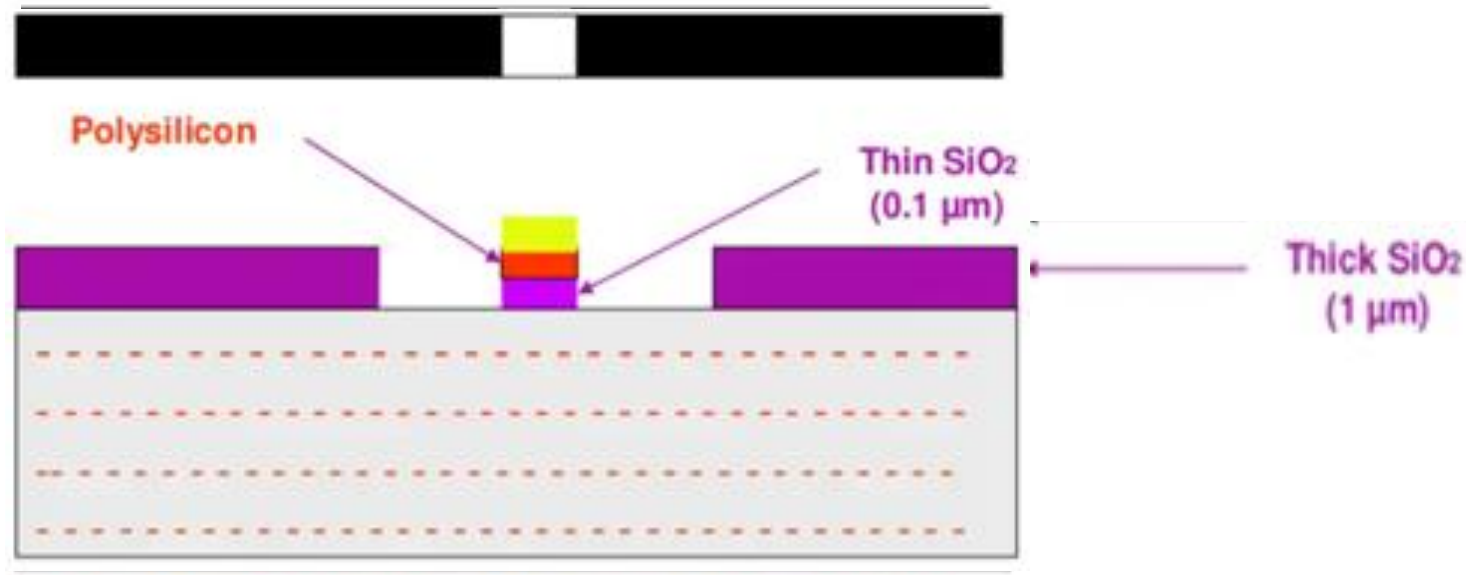
Step2: Depositing Polysilicon Layer

Step3: Growing of Photoresist layer over polysilicon.

Step4: Photoresist exposed to UV Rays using MASK2

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Formation of POLY gate:

Step1: Depositing Thin SiO₂ layer

Step2: Depositing Polysilicon Layer

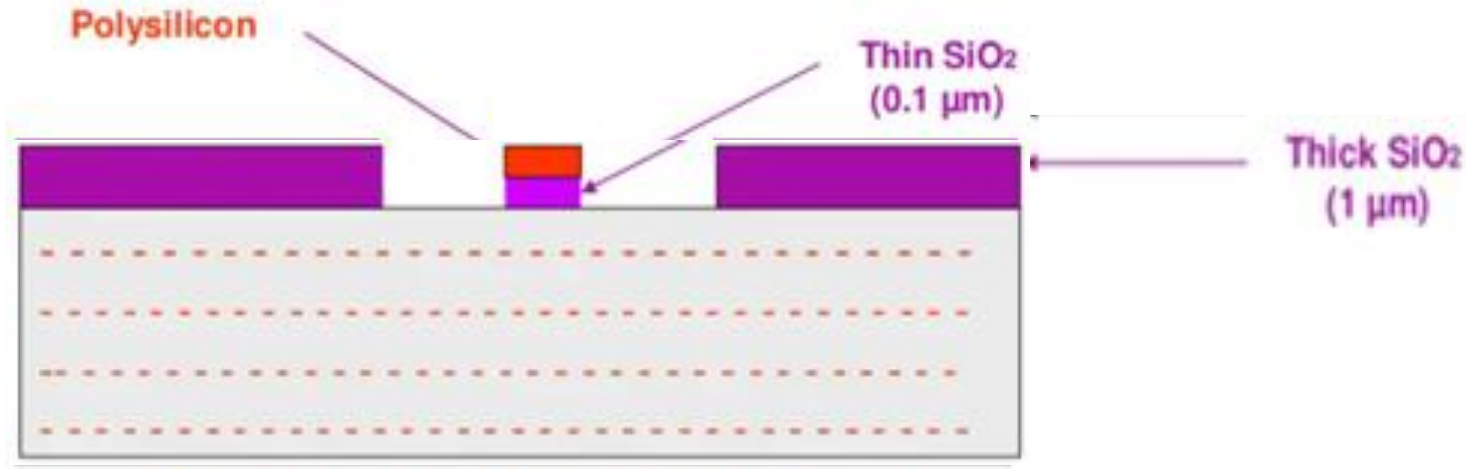
Step3: Growing of Photoresist layer over polysilicon.

Step4: Photoresist exposed to UV Rays using MASK2.

Step5: Etching of Soluble photoresist and SiO₂ below it.

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Formation of POLY gate:

Step1: Depositing Thin SiO₂ layer

Step2: Depositing Polysilicon Layer

Step3: Growing of Photoresist layer over polysilicon.

Step4: Photoresist exposed to UV Rays using MASK2.

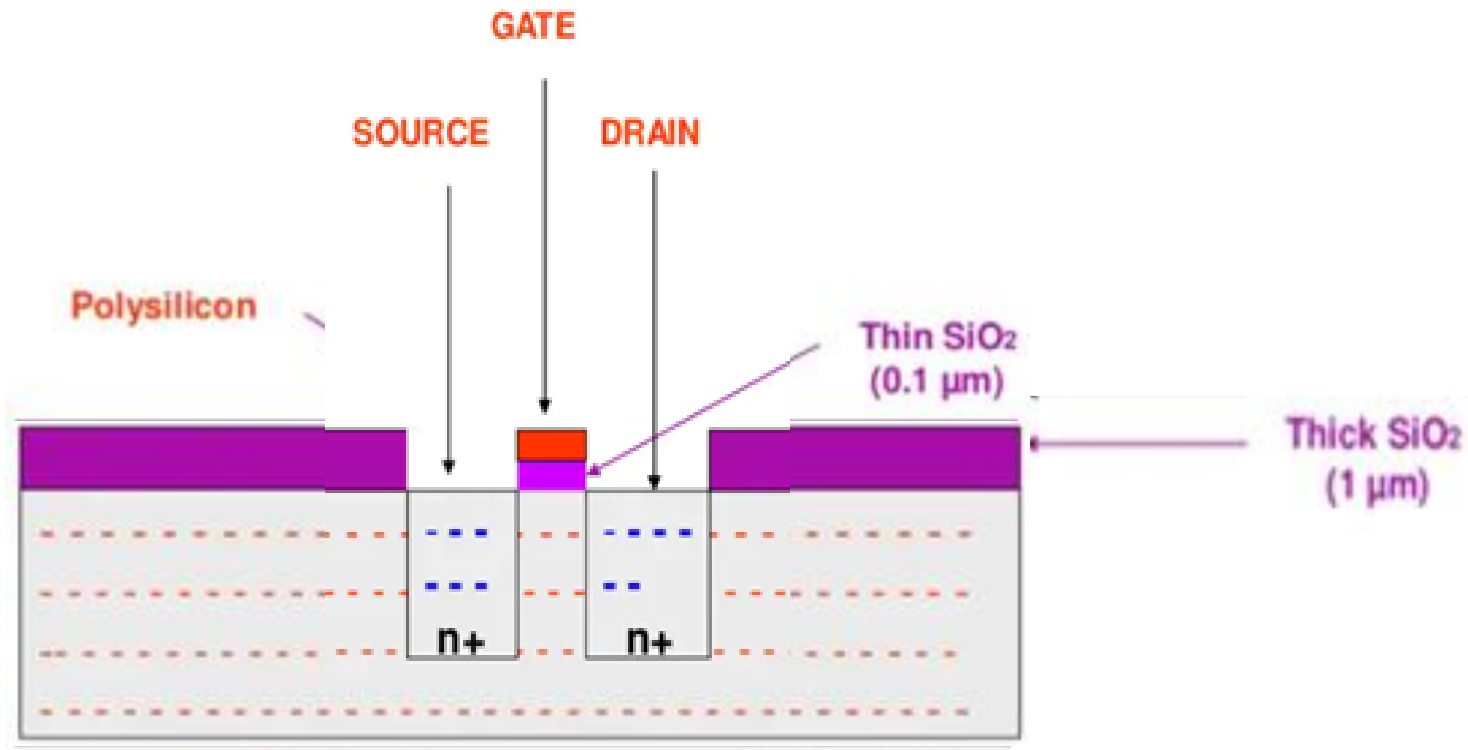
Step5: Etching of Soluble photoresist and SiO₂ below it.

Step6: Etching Unpolymerised photoresist material.

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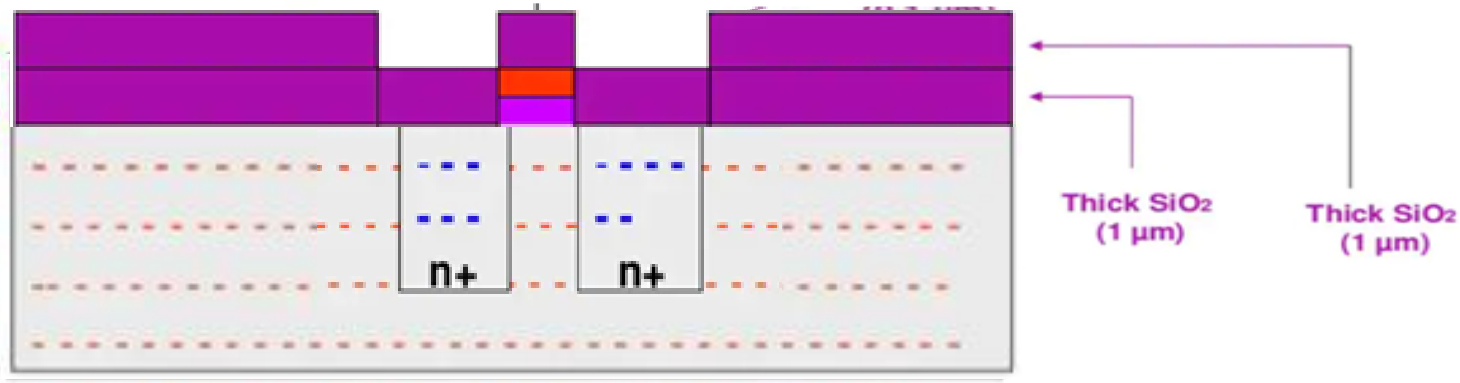
N+ doping to form Source and Drain terminal



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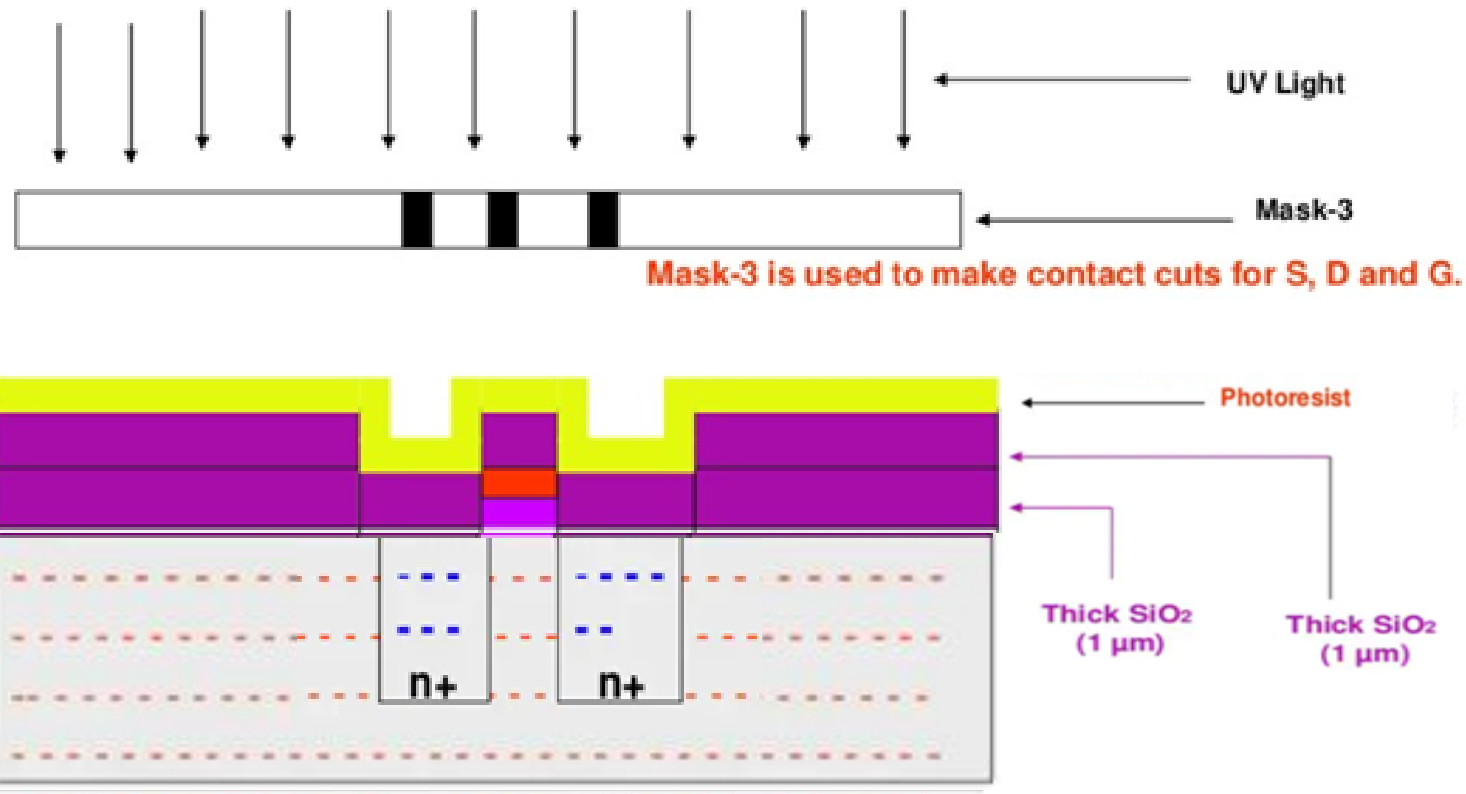
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Deposition of THICK SiO₂ layer



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Metallization:

Step1: Grow Photoresist Layer over SiO₂

Step2: Photoresist exposed to UV Rays using MASK3.

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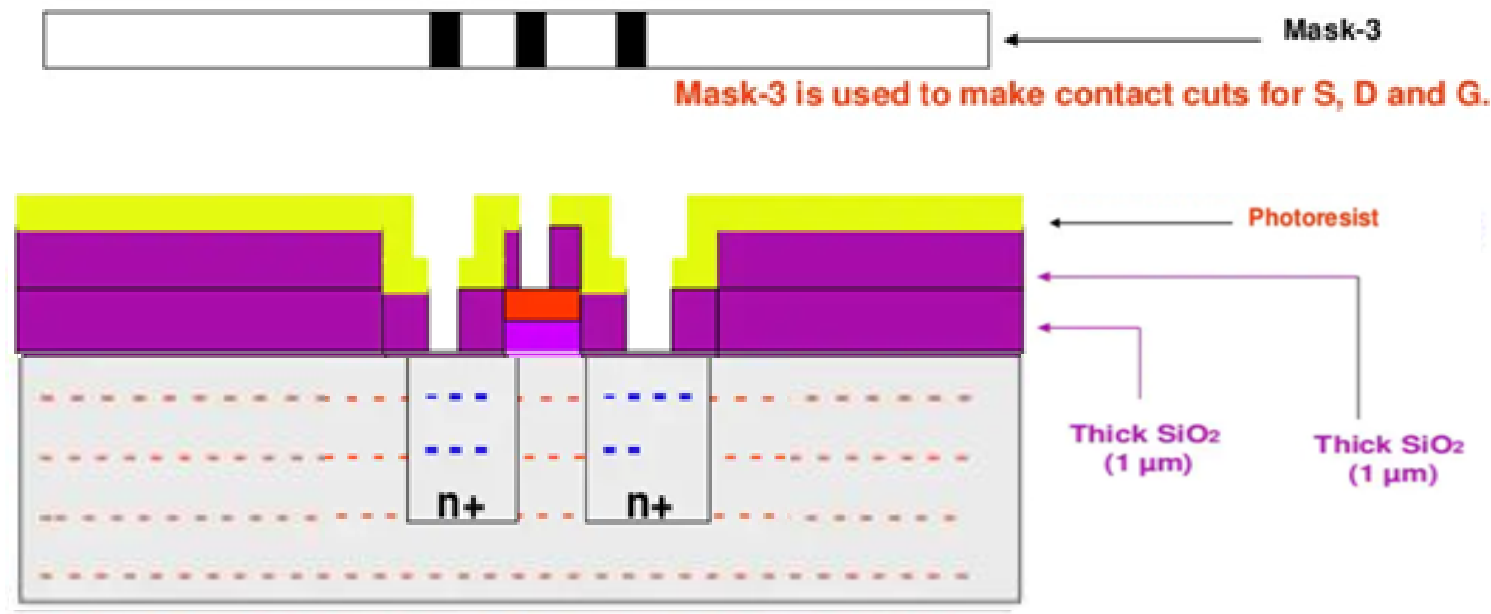
nMOS Fabrication Process

Metallization:

Step1: Grow Photoresist Layer over SiO₂

Step2: Photoresist exposed to UV Rays using MASK3.

Step3 Etching of Soluble photoresist and SiO₂ below it.



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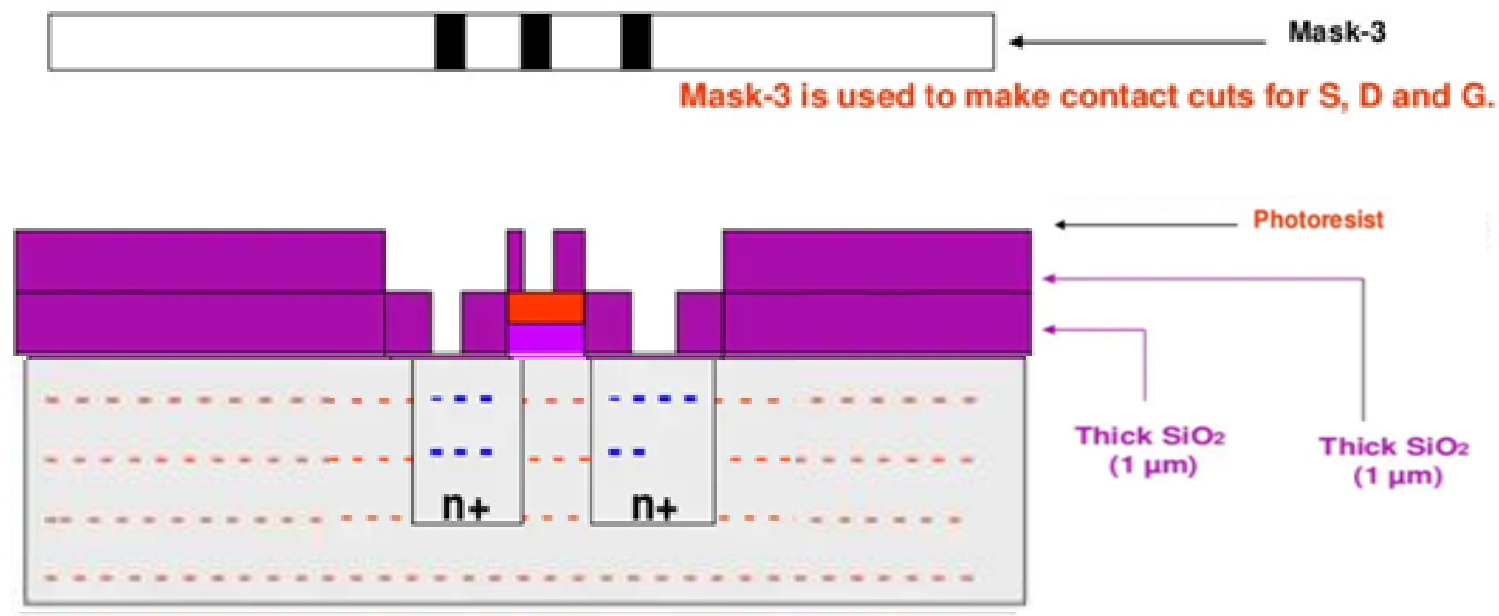
nMOS Fabrication Process

Metallization:

Step1: Grow Photoresist Layer over SiO₂

Step2: Photoresist exposed to UV Rays using MASK3.

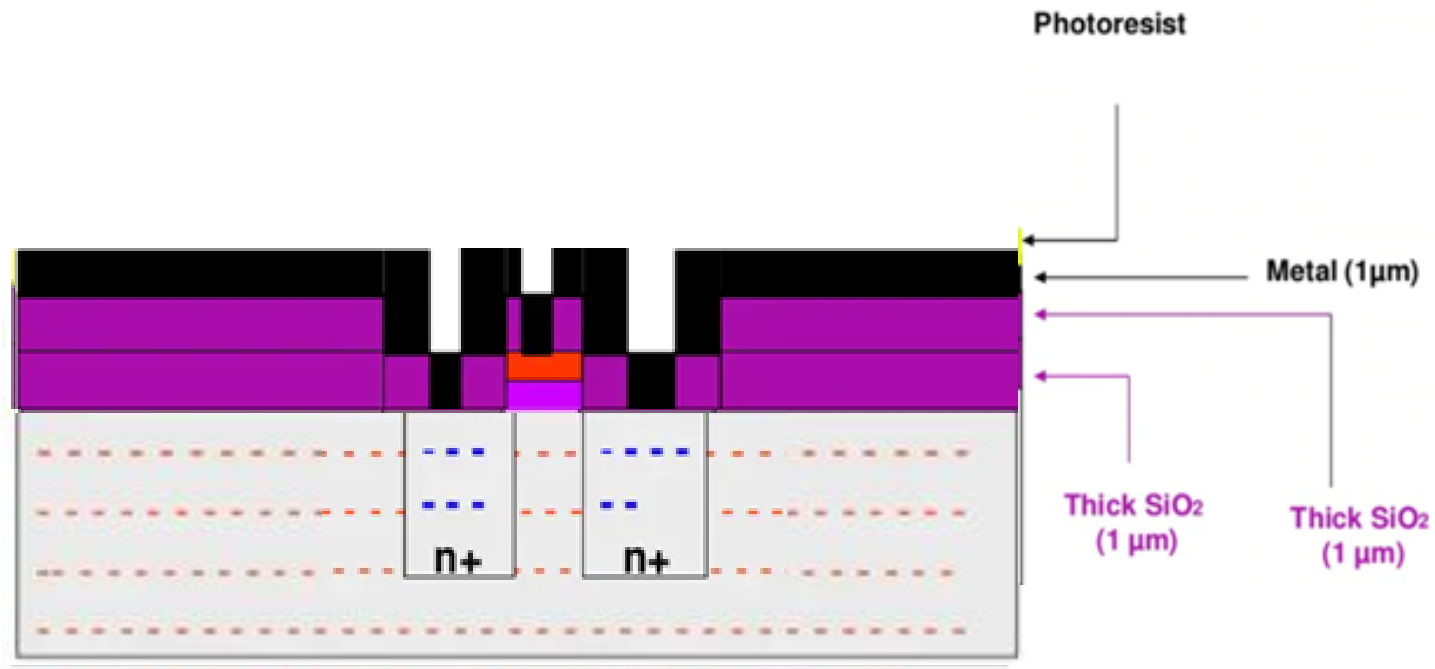
Step3 Etching of Soluble photoresist and SiO₂ below it.



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nMOS Fabrication Process

Deposition of Metal1 over the entire surface of chip (1u thickness)



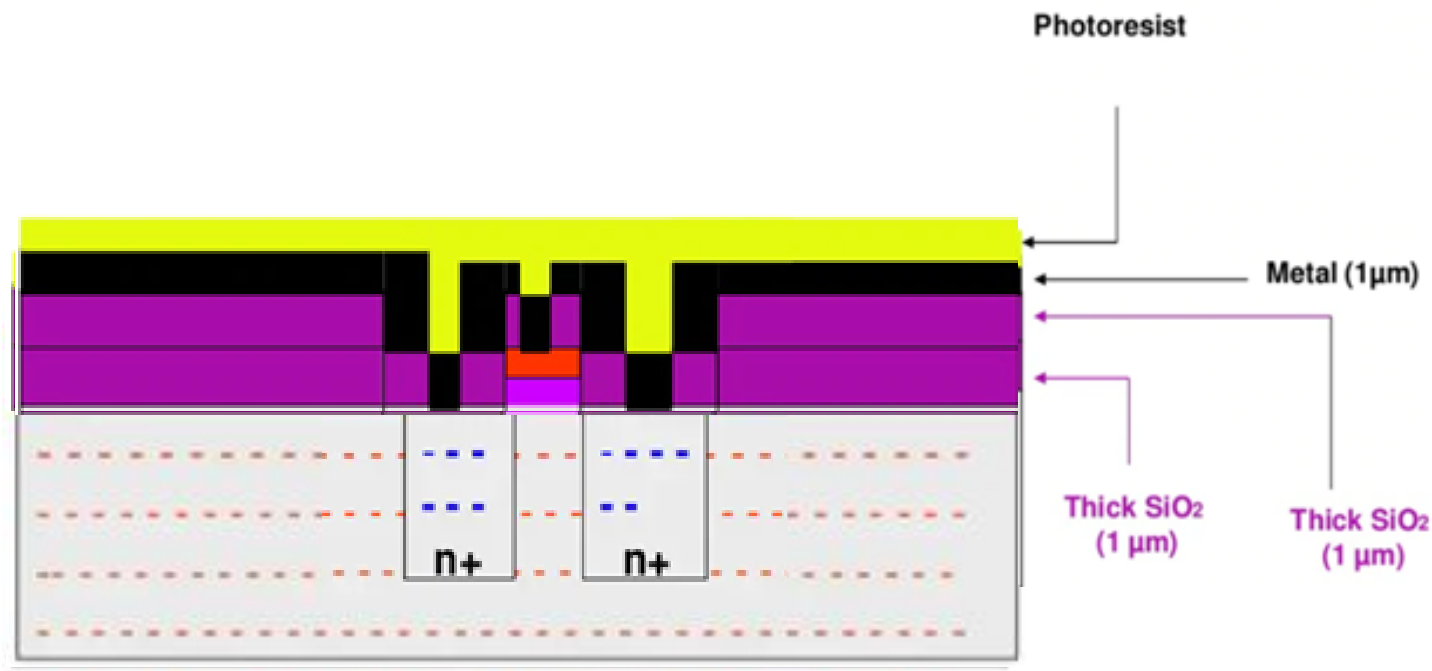
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Formation of Metal CONTACT:

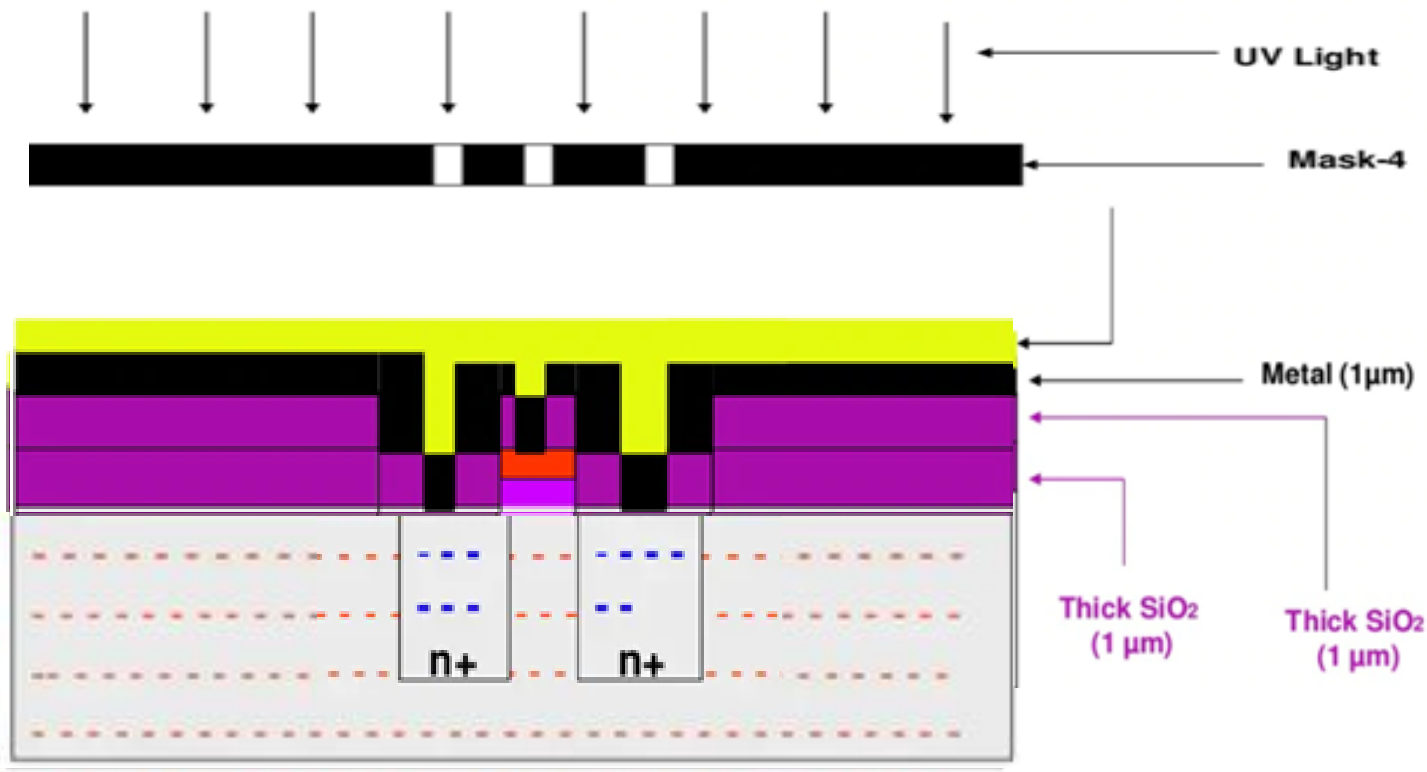
Step1: Deposition of Metal1 over the entire surface of chip (1u thickness)

Step2: Photoresist Deposition over Metal



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Formation of Metal CONTACT:

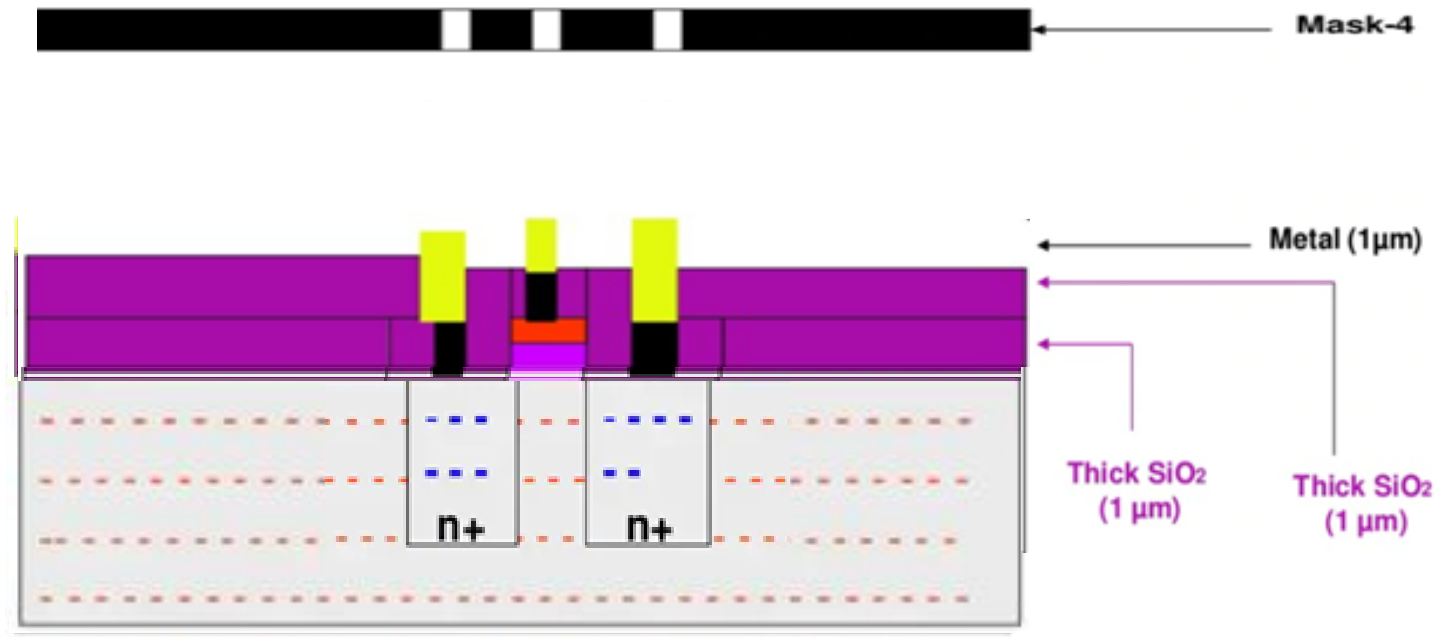
Step1: Deposition of Metal1 over the entire surface of chip (1 μ thickness)

Step2: Photoresist Deposition over Metal

Step3: Exposing photoresist to UV Rays using MASK4. This MASK exposes photoresist where contact need not be formed.

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Formation of Metal CONTACT:

Step1: Deposition of Metal1 over the entire surface of chip (1u thickness)

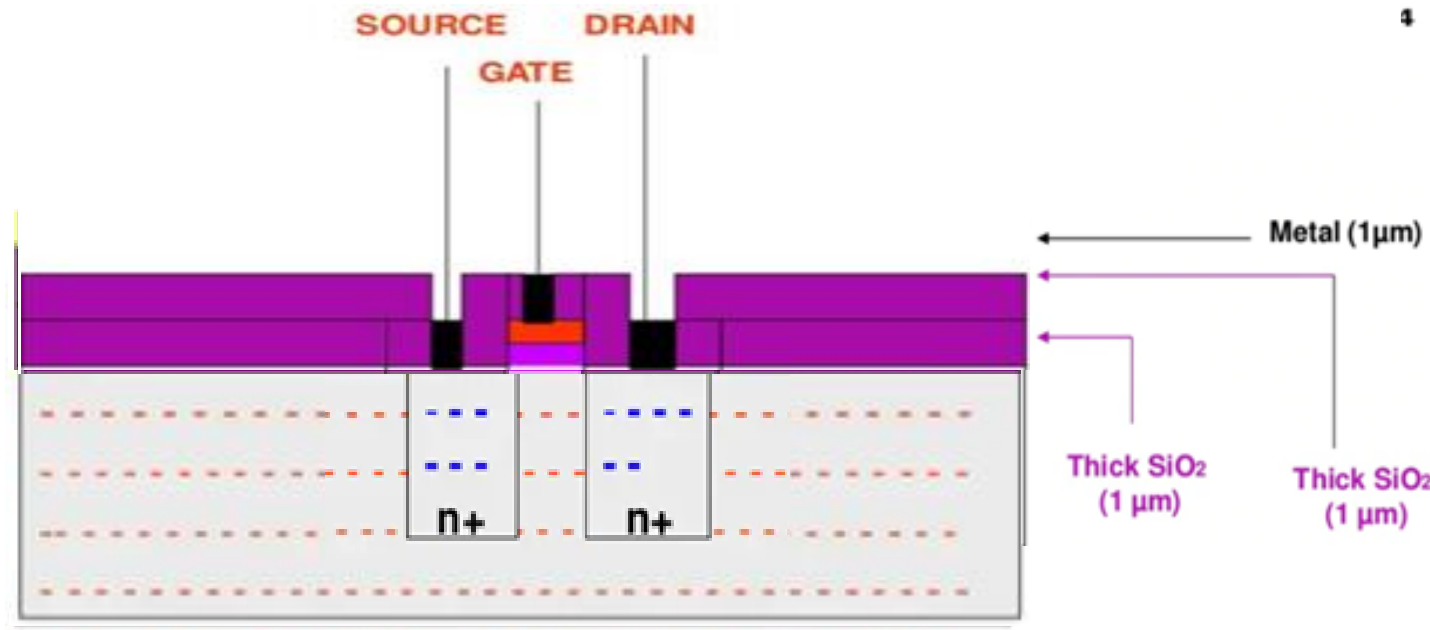
Step2: Photoresist Deposition over Metal

Step3: Exposing photoresist to UV Rays using MASK4. This MASK exposes photoresist where contact need not be formed.

Step4: Etching of polymerized photoresist layer

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Formation of Metal CONTACT:

Step1: Deposition of Metal1 over the entire surface of chip (1u thickness)

Step2: Photoresist Deposition over Metal

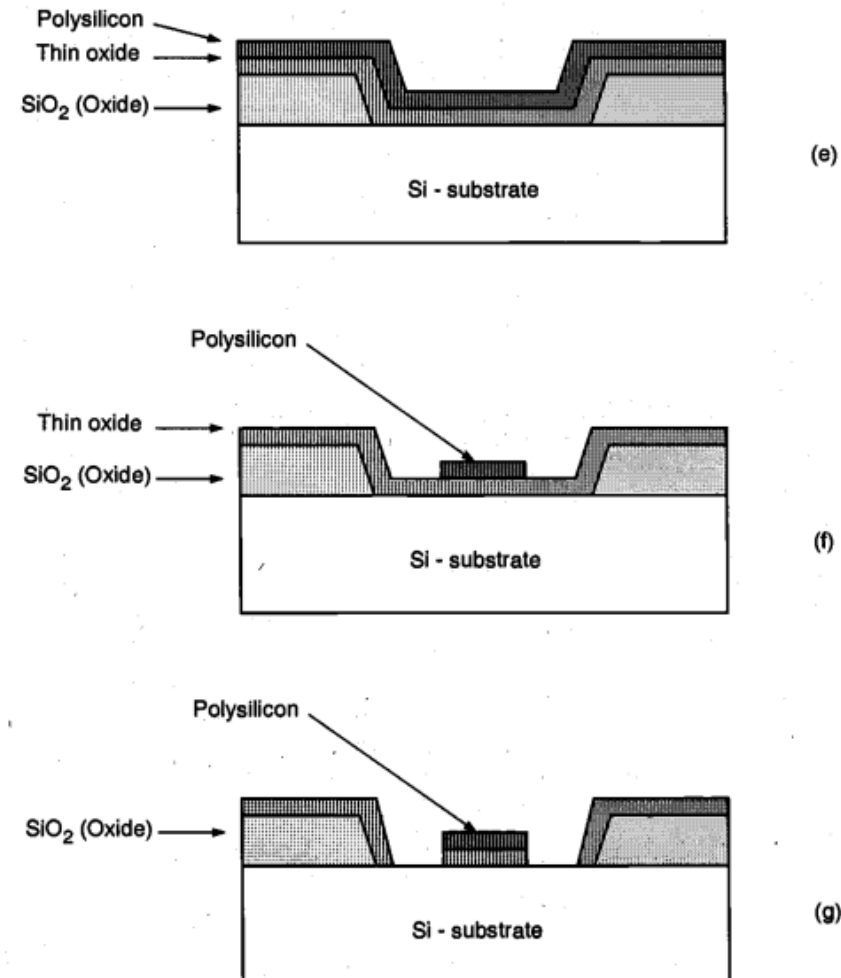
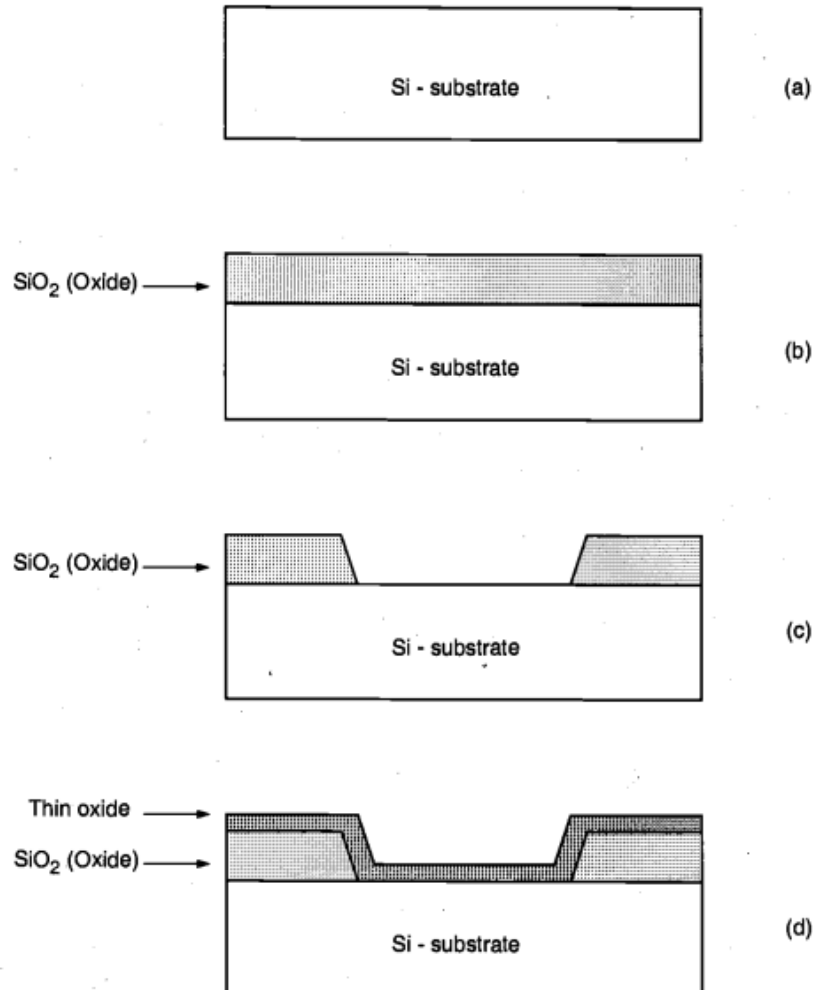
Step3: Exposing photoresist to UV Rays using MASK4. This MASK exposes photoresist where contact need not be formed.

Step4: Etching of polymerized photoresist layer

Step5: Etching un-polymerised photoresist

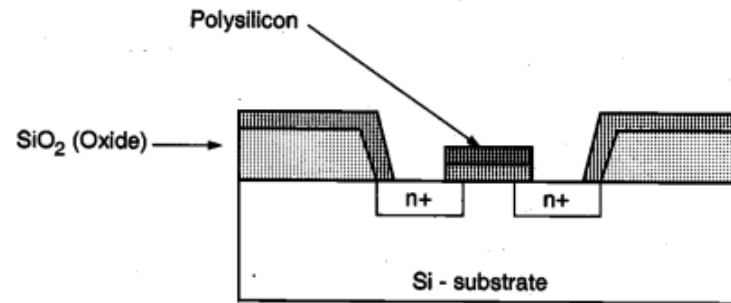
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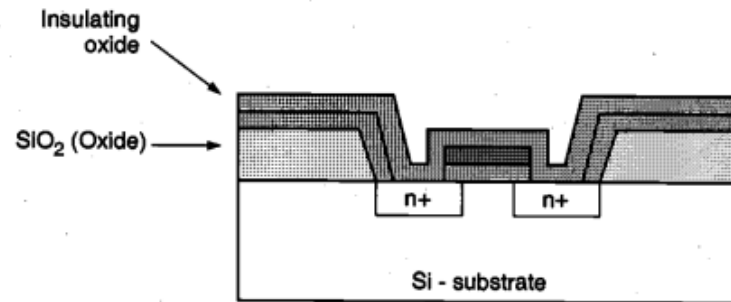


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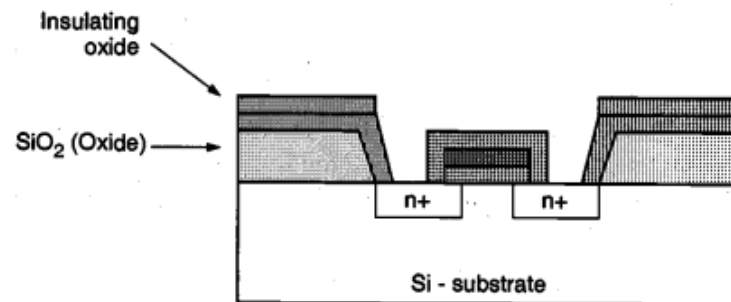
nMOS Fabrication Process



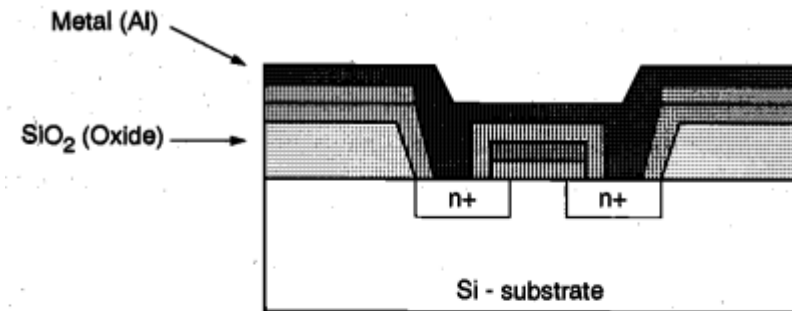
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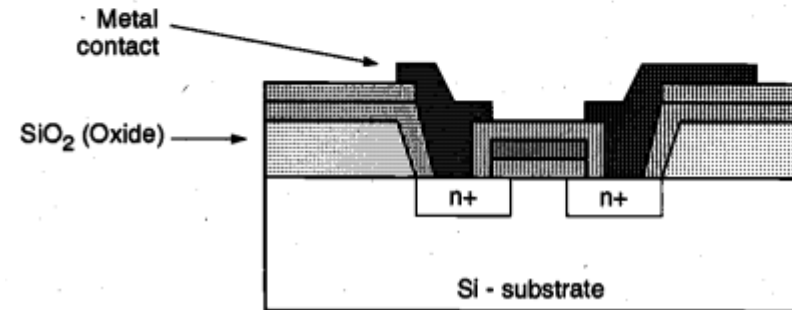
(i)



(j)



(k)



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THANK YOU

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